

A Unified Coherence Curve from Abiogenesis to 2200 CE: Stitching Biosphere, Civilization, and Simulated Futures

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Abstract

We present $C(t)$, a unified coherence curve spanning from the origin of life (3.8 Ga) to simulated futures (2200 CE), stitching together three measurement regimes: biosphere coherence $B(t)$ (3.8 Ga to 10,000 BCE, from paleontological data), civilizational coherence $K(t)$ (3000 BCE to 2020 CE, from historical proxies), and agent-based forward projection (2026 CE to 2200 CE, from multi-world civilization simulation).

The three regimes are connected via the **headroom formula**:

$$C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t))$$

which models civilization as filling the thermodynamic headroom above the biosphere floor, rather than replacing it. $B(t)$ is adjusted by Human Appropriation of Net Primary Productivity (HANPP) to reflect the energy that civilization co-opts from the biosphere.

The unified curve reveals three insights: (1) the biosphere maintained coherence > 0.85 through the pre-industrial era, and industrialization initially depressed total coherence via HANPP before civilizational contributions compensated; (2) the modern peak $C(t)(2020) = 0.94$ is the highest value in 3.8 billion years, achieved not by replacing biological coherence but by adding civilizational organization to it; and (3) forward projection under

baseline simulation parameters shows $C(t)$ stabilizing at 0.90–0.95 through 2200 CE, with sensitivity to governance quality and resource sustainability. **Keywords:** coherence, biosphere, civilization, unified index, HANPP, thermodynamic bridge, forward projection, deep time

Introduction

Measuring the organizational state of a planet—from the first replicating molecules to interplanetary civilization—requires a framework that spans radically different timescales, data sources, and phenomena. Previous work has addressed pieces of this span: biosphere coherence indices track deep-time organization from paleontological data [?]; composite civilizational indices (HDI, $K(t)$) measure modern and historical organization from economic and demographic proxies [?, ?]; and simulation models project civilizational futures under various policy assumptions [?].

No existing framework connects these three regimes into a single, continuous, thermodynamically grounded curve. The challenge is not merely mathematical (stitching different time series) but conceptual: what does it mean to compare a Cambrian biosphere’s organization to a 21st-century civilization’s? What physical quantity bridges the transition from biological to cultural coherence?

We propose that the bridge is **energy allocation**. Both the biosphere and civilization compete for the same planetary energy budget (solar insolation, chemical bonds). As civilization co-opts biosphere energy via land use, fossil extraction, and agriculture—quantified by Human Appropriation of Net Primary Productivity (HANPP) [?—the biosphere’s contribution to total coherence decreases while civilization’s increases. The *headroom formula* formalizes this: civilization fills the remaining organizational capacity above the biosphere floor, producing a unified index where the two contributions are additive rather than substitutive.

Methods

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Three Measurement Regimes

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$C(t)$ is computed in four phases:

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1. **Deep time** (3.8 Ga to 10,000 BCE): Pure $B(t)$, computed from
PBDB genus diversity, GEOCARB atmospheric data, ϵ_p isotopic
fractionation, Bambach ecospace functional groups, and background
extinction rates. Six dimensions aggregated via geometric mean [?].
2. **Holocene transition** (10,000 BCE to 1810 CE): $B(t)$ adjusted by
HANPP. Before recorded $K(t)$ data, the unified index is simply the
biosphere coherence reduced by human energy co-option: $C(t) =$
 $B(t) \times (1 - \text{HANPP}(t))$. HANPP rises from $\sim 0\%$ (pre-Neolithic) to
 $\sim 5\%$ (pre-industrial).
3. **Historical** (1810–2020 CE): The headroom formula applies. $K(t)$ is
computed from 35+ proxy variables across seven harmonies [?]. $B(t)$
continues to provide the biosphere floor, depressed by rising HANPP
(5% to 24%).
4. **Forward projection** (2026–2200 CE): An agent-based multi-world
civilization simulator [?] computes Eight Harmonies scores at each
epoch. These are passed to the headroom formula as the $K(t)$ contri-
bution, extending $C(t)$ into simulated futures.

The Headroom Formula

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The central equation:

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$$C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t)) \quad (1)$$

where $B_{\text{HANPP}}(t) = B(t) \times (1 - \text{HANPP}(t))$ is the biosphere coherence
adjusted for human energy appropriation.

This formulation has three properties: (1) when $K(t) = 0$ (pre-civilizational),
 $C(t) = B_{\text{HANPP}}(t)$ —pure biosphere; (2) when $K(t) = 1$ (maximal civiliza-

tional coherence), $C(t) = 1$ regardless of biosphere state; (3) for intermediate values, $K(t)$ fills the “headroom” between the biosphere floor and perfect coherence. Civilization cannot make $C(t)$ worse than the biosphere alone; it can only add to it.

HANPP Trajectory

Human Appropriation of Net Primary Productivity follows a sigmoidal trajectory calibrated from published estimates [?, ?]:

Era	HANPP (%)	B_{HANPP} (if $B = 0.95$)	
Pre-Neolithic	0	0.950	
Neolithic (3000 BCE)	1	0.941	
Roman Empire (0 CE)	3	0.922	58
Medieval (1000 CE)	4	0.912	
Industrial (1900 CE)	12	0.836	
Modern (2020 CE)	24	0.722	

Normalization Sensitivity

A systematic discrepancy exists between $K(t)$ values computed under different normalization strategies: min-max by century yields $K(2020) = 0.78$, while epoch-aware normalization yields $K(2020) = 0.91$ [?]. This propagates to $C(t)$: under the first, $C(2020) = 0.78 + 0.78 \times 0.22 = 0.95$; under the second, $C(2020) = 0.72 + 0.91 \times 0.28 = 0.97$. We report both and note that the qualitative shape of $C(t)$ is preserved across strategies.

Results

The Unified Curve

$C(t)$ spans 80 data points from 3.65 Ga to 2020 CE (Table 1).

Table 1. Selected points on the unified coherence curve.

Epoch	Regime	$C(t)$	B_{HANPP}	$K(t)$
3.6 Ga (Archean)	Biosphere	0.005	0.005	—
541 Ma (Cambrian)	Biosphere	0.127	0.127	—
66 Ma (K-Pg)	Biosphere	0.647	0.647	—
8000 BCE	Holocene	0.888	0.888	0.00
1000 CE	Holocene	0.908	0.908	0.00
1810 CE	Historical	0.895	0.895	0.00
1900 CE	Historical	0.860	0.836	0.14
1970 CE	Historical	0.857	0.779	0.34
2000 CE	Historical	0.774	0.722	0.11
2020 CE	Historical	0.939	0.722	0.78

Three Insights

Insight 1: Industrial depression and recovery. Between 1810 and 2000, HANPP rose from 5% to 24%, depressing B_{HANPP} from 0.90 to 0.72. Civilizational coherence $K(t)$ rose simultaneously from 0.0 to 0.78, but initially failed to compensate for the biosphere loss. The nadir at 2000 ($C = 0.77$) reflects a moment when civilization had depressed the biosphere substantially but had not yet filled the headroom. By 2020, $K(t) = 0.78$ fills 78% of the available headroom, pushing C back above 0.93.

Insight 2: The 2020 peak. $C(2020) = 0.94$ is the highest coherence value in 3.8 billion years. This does not mean 2020 is the “best” era—the index measures organizational complexity, not well-being or justice—but it means the combined biosphere-plus-civilization system has never been more structurally integrated.

Insight 3: The biosphere was more coherent alone. At 1000 CE, $C = 0.91$ with essentially zero $K(t)$ contribution. The biosphere *by itself*, before industrialization, achieved higher coherence than most of the industrial era. This quantifies the thermodynamic cost of the Anthropocene: civilization’s rise initially *reduced* total planetary coherence before eventually exceeding it.

Discussion

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Civilization as Thermodynamic Addition

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The headroom formula’s most important property is that it frames civilization as *additive*. Previous treatments of the Anthropocene tend to frame human impact as purely destructive—biosphere degradation, species loss, climate forcing. The $C(t)$ framework acknowledges this (via HANPP depression of the biosphere floor) while also accounting for what civilization *adds*: knowledge networks, governance structures, trade integration, reciprocal exchange. The question “is civilization good for the planet?” becomes quantifiable: $C(t)$ exceeds pre-civilizational $B(t)$ only when $K(t)$ fills more headroom than HANPP destroys.

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Connection to Consciousness

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The companion biosphere coherence paper [?] demonstrates that workspace (information capacity) was the rate-limiting dimension for consciousness emergence across 3.8 Ga. The SETI workspace filter paper [?] argues this constitutes a Great Filter. The $C(t)$ framework connects these findings to the present: civilizational coherence can be understood as the *continuation* of the workspace expansion that made consciousness possible. Human civilization is the mechanism by which the biosphere’s information capacity finally exceeded its biological ceiling—through culture, writing, computation, and global networks rather than through genome complexity alone.

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Limitations

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HANPP is coarse: Modern HANPP is estimated at 24% [?], but this aggregate masks regional variation (Europe > 40%, Amazon < 5%). Regional $C(t)$ decomposition is needed for policy applications.

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$K(t)$ normalization dependence: The 13-point divergence between $K(2020) = 0.78$ and $K(2020) = 0.91$ means absolute $C(t)$ values are normalization-dependent. Relative rankings are preserved.

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Forward projection is simulated: The 2026–2200 phase comes from
an agent-based model, not empirical data. Its predictions are conditional on
simulation assumptions (baseline governance, continued trade integration,
no major wars).

Single planet: $C(t)$ is calibrated to Earth. Whether the headroom
formula generalizes to other planet-civilization pairings is unknown.

Conclusion

We have produced a continuous coherence curve from the origin of life to the
present, and projected it into simulated futures, using a thermodynamically
grounded bridge between biosphere and civilization. The curve tells a story
of successive organizational innovations: from prokaryotic coherence (0.005)
to multicellular integration (0.13) to mammalian workspace expansion (0.65)
to civilizational overlay (0.94). Each transition built on—and in some cases
required catastrophic disruption of—the previous organizational regime.

The headroom formula ensures that this story is not one of replacement
but of addition. The biosphere is still there, beneath civilization, providing
the thermodynamic floor on which all human organization rests. The 24%
of NPP that we appropriate is the price of the organizational headroom we
fill. Whether we fill it wisely is not a question the index can answer, but it
is the question the index is designed to help us ask.

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